

# THE COMPLETE UBL MATH PACKET

## 1. Core Definition

$$\text{UBL} = \frac{P_{\text{domain}}}{P_{\text{Planck}}}$$

## 2. Planck Floor

$$P_{\text{Planck}} = \frac{c^5}{\hbar G^2}$$

Numerical:

$$P_{\text{Planck}} \approx 5.16 \times 10^{96} \text{ kg/m}^3$$

## 3. Vacuum Energy Density (from $\Lambda$ )

$$P_{\text{vac}} = \frac{\Lambda c^2}{8\pi G}$$

Numerical:

$$P_{\text{vac}} \approx 6 \times 10^{-2} \text{ kg/m}^3$$

## 4. UBL of Our Universe

$$\text{UBL}_{\text{universe}} = \frac{P_{\text{vac}}}{P_{\text{Planck}}}$$

Numerical:

$$\text{UBL}_{\text{universe}} \approx 10^{-123}$$

## 5. UBL Sub-Level Definition

For any physical scale with characteristic energy density  $P_{\text{sub}}$ :

$$\text{UBL}_{\text{sub}} = \frac{P_{\text{sub}}}{P_{\text{Planck}}}$$

## 6. UBL Log Scale

$$\log_{10}(\text{UBL}) = \log_{10}(P_{\text{sub}}) - \log_{10}(P_{\text{Planck}})$$

## 7. UBL Effective UV Cutoff

Domain-quietness-restricted cutoff:

$$\Lambda_{\text{eff}} = \text{UBL}^{1/4} \cdot M_{\text{Pl}}$$

Numerical for our universe:

$$\Lambda_{\text{eff}} \approx 10^{-31} M_{\text{Pl}}$$

## 8. Higgs Radiative Correction (Standard)

$$\delta m_H^2 \sim -\frac{y_t^2}{8\pi^2} \Lambda_{\text{UV}}^2$$

UBL-corrected:

$$\delta m_H^2 \sim -\frac{y_t^2}{8\pi^2} \Lambda_{\text{eff}}^2$$

## 9. Electroweak vs Gravity Coupling Ratio

$$\frac{G_{\text{EW}}}{G_{\text{grav}}} = \frac{\text{UBL}_{\text{EW}}}{\text{UBL}_{\text{universe}}}$$

Numerical:

$$\frac{G_{\text{EW}}}{G_{\text{grav}}} \approx \frac{10^{-40}}{10^{-123}} = 10^{83}$$

## 10. Zero-Point Energy Integral

Standard:

$$P_{\text{ZPE}} = \int_0^{\Lambda_{\text{UV}}} \frac{\hbar\omega}{2} g(\omega) d\omega$$

UBL-restricted:

$$P_{\text{ZPE}} = \int_0^{\Lambda_{\text{eff}}} \frac{\hbar\omega}{2} g(\omega) d\omega$$

## 11. Renormalization Scale Identification

$$\mu_{\text{phys}} = \Lambda_{\text{eff}} = \text{UBL}^{1/4} M_{\text{Pl}}$$

## 12. QCD Confinement Scale Relation

$$\Lambda_{\text{QCD}} \sim M_{\text{Pl}} \cdot 10^{-1}$$

UBL mapping:

$$\text{UBL}_{\text{QCD}} = \frac{P_{\text{QCD}}}{P_{\text{Planck}}} \approx 10^{-65}$$

## 13. Decoherence Rate Scaling

$$\Gamma_{\text{decohere}} \sim \left( \frac{m}{m_{\text{Pl}}} \right)^2 \cdot \text{UBL}^{-1} \cdot \omega_{\text{Pl}}$$

## 14. Gravity vs EM Force Ratio

$$\frac{F_{\text{grav}}}{F_{\text{EM}}} = \frac{Gm_p^2}{k_e e^2}$$

UBL interpretation:

$$\frac{F_{\text{grav}}}{F_{\text{EM}}} \sim \frac{\text{UBL}_{\text{universe}}}{\text{UBL}_{\text{atomic}}}$$

Numerical:

$$\frac{F_{\text{grav}}}{F_{\text{EM}}} \approx 10^{-37}$$

## 15. UBL Mode-Suppression Condition

A vacuum mode of frequency  $\omega$  is accessible only if:

$$\frac{\hbar\omega}{V} \leq \text{UBL} \cdot P_{\text{Planck}}$$

Equivalently:

$$\omega \leq \omega_{\text{max}} = \text{UBL}^{1/4} \omega_{\text{Pl}}$$

## 16. Domain-Level Vacuum Stress-Energy

$$T_{\text{vac}}^{\mu\nu} = \text{UBL} \cdot T_{\text{Planck}}^{\mu\nu}$$

### 17. UBL Step Differences Between Scales

For any two scales  $a$  and  $b$ :

$$\Delta_{\text{UBL}} = \log_{10} \left( \frac{P_a}{P_b} \right)$$

### 18. UBL-Driven Running Cutoff

$$\Lambda(\text{scale}) = \Lambda_{\text{eff}} \cdot f(\text{RGE})$$

Where  $f(\text{RGE})$  is the usual renormalization-group flow.

### 19. UBL-Restricted Virtual Mode Density

$$\rho_{\text{modes}} = \rho_{\text{Planck}} \cdot \text{UBL}$$

### 20. UBL-Restricted Vacuum Fluctuation Amplitude

$$\langle \delta\phi^2 \rangle_{\text{UBL}} = \text{UBL} \cdot \langle \delta\phi^2 \rangle_{\text{Planck}}$$